30-Point Kubernetes Cluster Health Checklist

30 checks. Zero theory.
Built for teams who own uptime and cost.

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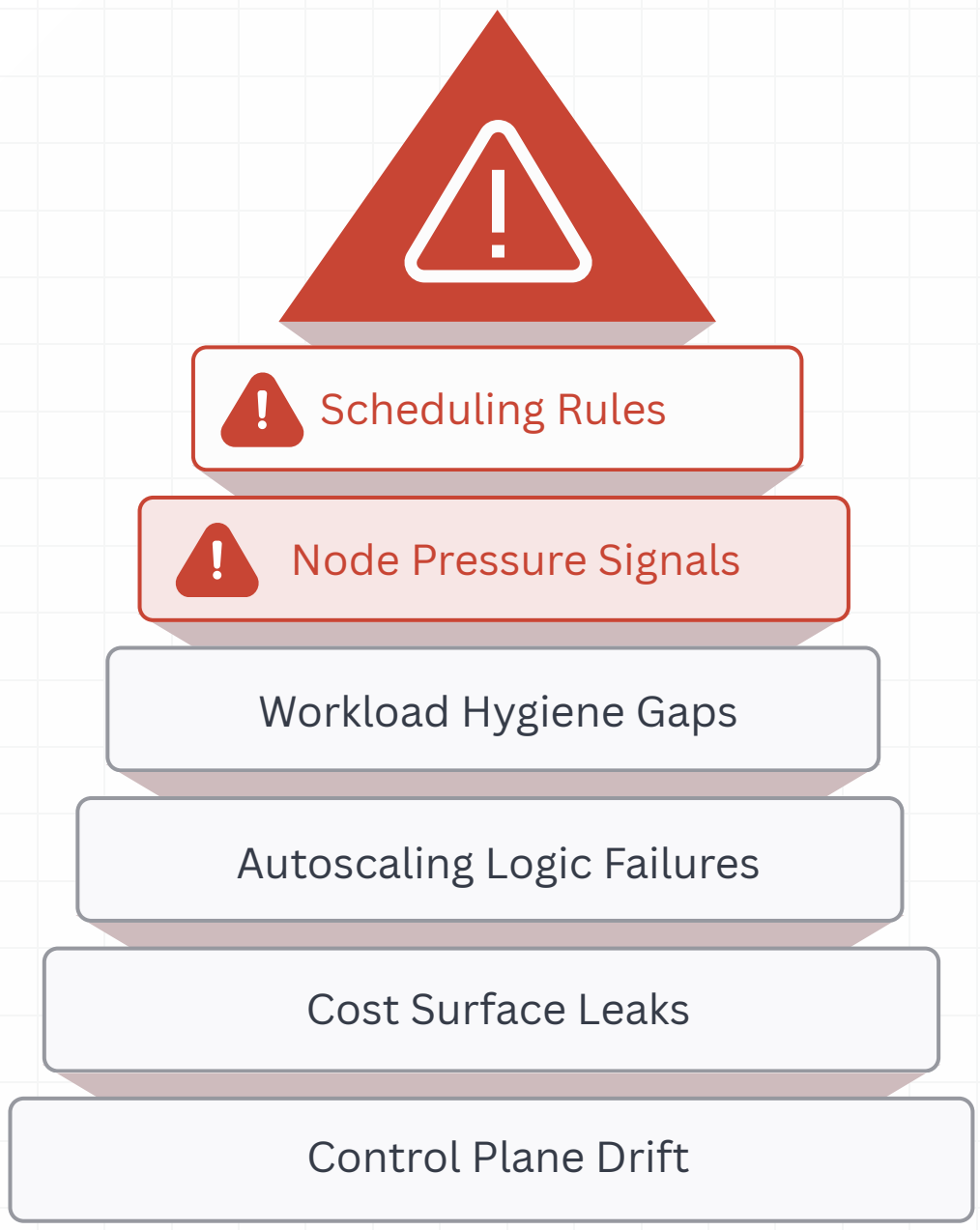
How This Checklist Works

Why Monthly?

Clusters don’t explode randomly.
They decay slowly — then fail loud at 3am.

These 30 checks are built to catch silent failures
before they wreck your uptime or spike your cloud bill.



Kubernetes Fails in Layers

Each section in this checklist maps to one of these layers.

Top layers = most likely to cause real incidents.

How to Use It

- ☑ Run monthly (Infra360 default cadence)
- ☑ Copy commands directly
- ☑ Flag check owner

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01—06: Scheduling & Survival

(Pods that live vs die)

No.	Mechanism	Exact Check
01	Pods without resource requests	kubectl get pod -A -o=jsonpath='{range .items[]}{.metadata.namespace}/{.metadata.name}{"\t"}{.spec.containers[].resources}{"\n"}{end}' | grep -v requests
02	QoS class distribution (BestEffort present in prod = red flag)	kubectl get pod -A -o custom-columns=NS:.metadata.namespace,NAME:.metadata.name,QOS:.status.qosClass
03	Requests > node allocatable	kubectl describe node | grep -A5 Allocated
04	Critical workloads not Guaranteed QoS	same QoS command + service tiering
05	Over-replication “just in case”	kubectl get deploy -A -o custom-columns=NS:.metadata.namespace,NAME:.metadata.name,REPLICAS:.spec.replicas
06	DaemonSets not included in capacity math	kubectl get ds -A -o custom-columns=NS:.metadata.namespace,NAME:.metadata.name,REQ:.spec.template.spec.containers[*].resources.requests

07—11: Node Reality Checks

(What kubelet is screaming quietly)

No.	Mechanism	Exact Check
07	MemoryPressure / DiskPressure ignored	`kubectl describe node | egrep "MemoryPressure|DiskPressure|PIDPressure"`
08	Evictions happening quietly	`kubectl describe node | grep -i eviction`
09	ImageFS filling up	`df -h /var/lib/containerd /var/lib/docker`
10	PID exhaustion risk	`kubectl describe node | grep -i pid`
11	Uneven pod distribution across nodes	`kubectl get pod -A -o wide`

12—16: Workload Hygiene

(Things that rot slowly)

No.	Mechanism	Exact Check
12	CrashLoopBackOff masking startup/config bugs	kubectl get pod -A | grep CrashLoopBackOff kubectl logs <pod> --previous
13	InitContainers failing but ignored	kubectl get pod -A | grep Init
14	Old ReplicaSets piling up	kubectl get rs -A | wc -l Fix: revisionHistoryLimit
15	Deployments with no liveness/readiness probes	kubectl get deploy -A -o=jsonpath='{range .items[]}{.metadata.namespace}/{.metadata.name}{"\t"}{.spec.template.spec.containers[].livenessProbe}{"\n"}{end}'
16	Probes too aggressive (self-DDOS)	probe configs vs startup time

17—20: Autoscaling Illusions

No.	Mechanism	Exact Check
17	HPA scaling on CPU for non-CPU workloads	`kubectl get hpa -A`
18	HPA maxReplicas never hit (means it's useless)	`kubectl describe hpa`
19	VPA running without guardrails	`kubectl get vpa -A`
20	Cluster Autoscaler fighting PDBs	`kubectl get pdb -A`

21—25: Networking & Cloud Cost Bleeds

No.	Mechanism	Exact Check
21	Zombie namespaces	`kubectl get ns --sort-by=.metadata.creationTimestamp`
22	Orphaned LoadBalancer services	`kubectl get svc -A | grep LoadBalancer`
23	Idle node pools kept alive by 1 pod	`kubectl get pod -A -o wide`
24	Ingress controllers over-provisioned	`kubectl get deploy -n ingress-nginx`
25	Unused PVCs still attached	`kubectl get pvc -A`

26—30: Control Plane & Signal Hygiene

No.	Mechanism	Exact Check
26	API server event spam ignored	kubectl get events -A --sort-by=.lastTimestamp | tail -50
27	Excessive watch/list traffic (client abuse)	API server metrics / audit logs
28	RBAC drift (old service accounts still live)	kubectl get sa -A
29	Secrets never rotated	kubectl get secret -A --sort-by=.metadata.creationTimestamp
30	No human ownership per namespace	kubectl get ns --show-labels Look for owner=